8. Which of the following equalities is satisfied for all real numbers x and y?

A. $3^{x+y}3^{x-y} = 3^{x^2-y^2}$

B. $3^{x+y}3^{x-y} = (3^x)^2$

C. $3^{x+y}3^{x-y} = 3^{x^2} - 3^{y^2}$

D. $3^{x+y}3^{x-y} = 3^{x^2}$

E. $3^{x+y}3^{x-y} = 3^x(3^y 3^{-y})$